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Special Section:
Advancing process representation in hydrologic models: Integrating new concepts, knowledge, and data

Estimating Vegetation Greening Influences on Runoff Signatures Using a Log-Based Weighted Ensemble Method

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Key Points:

- Isolate vegetation greening impact on runoff signatures by comparing modeling experiments applying a log-based weighted ensemble approach
- Increasing leaf area index decreased flow magnitude, increased low flow duration, and resulted in little change in flow dynamic signatures
- Flow frequency and duration signatures showed a greater response than magnitude signatures

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Vegetation greening profoundly impacts the water cycle, and recent concerns about greening impacts have focused on various hydrological cycle components. However, the impacts of greening on catchment runoff signatures reflecting magnitude, low/high flow frequency, low/high flow duration and flow dynamics remain poorly understood. To properly simulate these runoff signatures, we use five modified hydrological models incorporating vegetation dynamics and further derive three ensemble approaches to obtain eight runoff time series outputs in a major tributary of the Yellow River Basin. Multiple validations suggest that the log-based weighted ensemble (LWE) approach is robust for depicting the impact of greening on selected runoff signatures. This is especially true for the low flow part of the runoff time series and the overall performance of the selected signatures since LWE explicitly reduces the low flow bias. With this approach, five experiments were designed to isolate the impact of vegetation greening on runoff signatures, and the comparisons among the experiments indicate that greening noticeably decreases runoff magnitude, increases low flow frequency/duration and decreases high flow frequency/duration signatures. However, greening has little influence on runoff dynamic signatures. Each percent increase in leaf area index results in (a) changes of $-0.2 \pm 0.1\%$ for magnitude signatures; (b) changes of $-0.34 \pm 0.30\%$ and $0.56 \pm 0.28\%$ with wide ranges for annual high flow days and annual low flow days, respectively; and (c) marginal change on flow dynamic signatures. This study provides new insights by disentangling greening impacts on various runoff signatures using a trade-off ensemble method.

1. Introduction

Continuous and widespread vegetation change (including both leaf area index [LAI] increase and land cover [LC] conversion, etc.) has been reported in most regions of the world (Chen et al., 2019; Mao et al., 2016; Mastrotheodoros et al., 2020; Myers-Smith et al., 2020; Zhu et al., 2016). From 2000 to 2017, one-third of the global vegetated area showed an increasing in greening, which was especially obvious in China and India (Chen et al., 2019). The increase in vegetation greenness has brought substantial impacts on climate and the water cycle, including land surface temperature (Chen et al., 2020; Zeng et al., 2017), evapotranspiration (Ma et al., 2021; Meng et al., 2020; Zeng et al., 2018; Zhang et al., 2021), soil water (Liu et al., 2022; Zhang, Wang, et al., 2020) and runoff (Liu et al., 2016, Liu, Liang, et al., 2020; Ukkola et al., 2016). Moreover, several studies projected an increased LAI across the majority of China due to warming and elevated CO_2 in the future (Liu, Wang, et al., 2020; Ma, Zhao, et al., 2019; Yuan et al., 2021; Zhao et al., 2020).

The Yellow River Basin, a mostly arid and semiarid basin in China, has shown a notable increase in LAI since 1999 when the Grain-To-Green Project was implemented (Wang, Gao, et al., 2017; Wang et al., 2019). Although climate change is the main driver of change in runoff (Liu et al., 2018), the LAI increase led to more runoff reduction in the Yellow River Basin in the 2000s (Zhai & Tao, 2021). Previous studies (Bai et al., 2019, 2020; Zeng et al., 2018; Zhang et al., 2017) showed that greening exacerbated water scarcity in the water-limited regions in China with increased evapotranspiration and decreased annual runoff. Furthermore, runoff is the most critical factor to provide water for human living, irrigated agriculture and ecological health. Thus, the concern of this study is how greening may impact runoff, especially for arid and semiarid regions (e.g., Yellow River Basin).

Numerous studies have investigated greening-induced changes in runoff magnitude based on either experimental approaches or hydrological modeling. The impacts of greening on runoff magnitude have been estimated by time series analysis or by comparing runoff under greening and nongreening experiments, including (a) time series analysis within the same basin (b) paired catchments (i.e., similar in most attributes other than vegetation),

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comparisons (Bosch & Hewlett, 1982; Brown et al., 2005; Hawthorne et al., 2013), and (c) process-based hydrological modeling of pregreening and postgreening changes (Tesemma et al., 2015; Zhang et al., 2018) or hydrological models with original and modified static vegetation data, respectively (Bai et al., 2020; Das et al., 2018; Meng et al., 2020; Mohaideen & Varija, 2018). Table 1 summarizes recent papers on vegetation influence on runoff change. Some obvious challenges can be seen from the papers in Table 1. For the time series analysis method, the main challenge is to collect long-term vegetation and runoff data. For the paired catchment method, there are always difficulties in selecting two similar catchments. For the hydrological modeling method, the models may be overoptimized to some extent, ignoring other runoff processes. Moreover, as shown in Table 1, most of these previous studies focus on runoff volume change, drawing a common conclusion that greening generally reduces runoff magnitude. The decreasing (increasing) flow magnitude may result in an increasing (decreasing) number of low flow days. However, this has not been fully discussed, and influences on low/high flow frequency/duration or flow dynamics have not been comprehensively quantified. To further detect the influ-

Table 1
A Summary of Several Recent Studies Referring to the Impacts of Vegetation Change on Runoff

Focused aspects	Approaches	Vegetation change variable	Model/method	Experiment	Study area	Major conclusion	Reference
Volume change	Time series analysis method	LUCC	Improved model of the Budyko elasticity model	Split the attribution of vegetation and climate change to runoff change	Yellow River Basin, China	Vegetation change (mainly anthropogenic vegetation coverage increase) accounted for 92% of the decline in runoff	(Li, Shi, et al., 2020)
	Time series analysis method	Vegetation increase	Budyko hypothesis	Attributed the runoff change between before and after the breakpoint	Hai River Basin, China	Impacts of land use/cover change were accountable for the runoff decrease by 73.1% on average.	(Xu et al., 2014)
	Time series analysis method	Agricultural land use change	Trend analysis and Elastic analysis	Quantified the elasticity of base flow to climate variability and agricultural land use change.	Missouri River Basin (MORB)	Agricultural land use increases mostly led to a 0%–5.7% decrease in annual base flow in the basin	(Ahiablame et al., 2017)
	Paired catchment	LAI, Normalized differential vegetation index (NDVI),	Principal component analysis, step regression analysis, and sensitivity analysis	Regression and sensitivity analysis	Hai River Basin, China	The runoff capacity of Hai River Basin was mainly controlled by vegetation types and soil texture	(Zhang, Liu, et al., 2020)
	Modeling	LAI change	Budyko-Choudhury-Porporato Model linked with Guswa's Rooting Depth Model	1. Past scenario 2. Future scenario 3. Attribution Analysis of Hydrological Partitioning	75 Catchments in China	Increased LAI has led to substantial streamflow reductions over the past three decades (on average ~8.5% in 1990s and ~11.7% in 2000s compared to the 1980s baseline	(Zhang, Yang, et al., 2018)

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Table 1
Continued

Focused aspects	Approaches	Vegetation change variable	Model/method	Experiment	Study area	Major conclusion	Reference
	Modeling	LAI change	The variable infiltration capacity (VIC) model combined with LAI estimating model	1. The "Millennium Drought" (1997–2009) relative to the period 1983–1995, 2. Two future periods (2021–2050 and 2071–2100) and two emission scenarios compared with the baseline (1981–2010)	13 catchments in Australia	Including the effects of changes in LAI in future projections of streamflow is important.	(Tesemma et al., 2015)
	Modeling	–	Empirical model and simple water balance model	1. Baseline period (1961–1986) 2. Changeable period (1987–2000);	Yiluo River Basin, China	The vegetation changes were the main cause for change of annual runoff from baseline period to changeable period	(Liu et al., 2009)
	Modeling	LAI change	The variable infiltration capacity (VIC) macro scale hydrologic model	1. Precipitation-detrend simulation scenario 2. Temperature-detrend simulation scenario 3. LAI-detrend scenario 4. Baseline simulation	Three-North region (TNR) of China	Precipitation change had a more significant influence on the ET, soil moisture and runoff than vegetation growth at the regional scale.	(Meng et al., 2020)
	Modeling	LAI change	A process-based ecosystem model	1. Original LAI 2. Detrended LAI	China	Vegetation greening increased annual ET and decreased water yield	(Liu et al., 2016)
	Modeling	LAI change	The Variable Infiltration Capacity (VIC) model	Factor-controlled simulation experiments	Conterminous United States (CONUS)	Changing LAI is the second most important factor in controlling variability in ET and SM, particularly over the cool season.	(Liu et al., 2018)
	Modeling	NDVI, albedo	abcd_PML model	1. The original data inputs 2. Detrended NDVI and albedo data inputs	China	Greening contribution to annual ET and water yield is 0.25 and −0.16 mm/yr, respectively.	(Bai et al., 2020)

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Table 1
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Focused aspects	Approaches	Vegetation change variable	Model/method	Experiment	Study area	Major conclusion	Reference
	Modeling	LAI change	Community Land Model (CLMv4.5)	1. Dynamic vegetation inputs 2. Climatological vegetation inputs	Minnesota (USA)	The average magnitude difference of groundwater recharge was 21.1% in response to only a 4% difference in LAI inputs	(Anurag et al., 2021)
	Modeling	IGBP Land use	Water yield response model	1. Pre-forestation periods 2. Postforestation periods	China	Forestation of China would have the potential to reduce streamflow by 50–150 mm/yr or 20%–40% across most parts of the nation.	(Sun et al., 2006)
	Modeling	Land use and land cover change (LULCC)	The variable infiltration capacity (VIC) macro scale hydrologic model	The model simulations were performed for each LULC scenario by keeping climate data (of the year 2005) invariant	Five basins in eastern India	Deforestation, urbanization, cropland expansions led to reduced canopy cover for interception and transpiration that in turn contributed to overall decrease in ET and increase in runoff and base flow	(Das et al., 2018)
	Modeling	Land use	An Improved Soil and Water Assessment Tool (SWAT-T) model	1. Deforestation 2. Urbanization/ deforestation 3. Afforestation 4. Urbanization	The North Johnstone River catchment, Australia	Urbanization increased surface runoff and decreased lateral runoff and groundwater. Afforestation decreased surface runoff and increased evapotranspiration	(Zhang, Wang, et al., 2020)
	Paired catchments; hydrological Modeling; Time series analysis method; etc.	Forest cover change	–	–	312 Watersheds worldwide in the literature	Annual runoff in water-limited watersheds is more sensitive to forest cover change than in energy-limited watersheds	(Zhang et al., 2017)

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Table 1
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Focused aspects	Approaches	Vegetation change variable	Model/method	Experiment	Study area	Major conclusion	Reference
Volume, seasonal and regime change	Paired catchment	Vegetation type conversion	-	1. Afforestation experiments 2. Regrowth experiments 3. Deforestation experiments 4. Forest conversion experiments	166 Paired catchments of Australia, India, New Zealand, South Africa, USA, UK	This review highlights the lack of information available in the literature for making quantitative generalizations about the impacts of vegetation changes on seasonal yield and flow regime.	(Brown et al., 2005)
Runoff signature change (magnitude, frequency, duration and dynamic)	Modeling (improved hydrological models with better representation of ET)	LAI change and LC change	Eight approaches (five improved hydrological models and three ensemble methods)	1. Actual condition (observed LAI and LC inputs), 2. LAI-detrended condition (observed LC inputs, detrended LAI), 3. LC-detrended condition (observed LAI inputs, LC for the first year); and 4. LAI-LC-detrended condition (detrended LAI inputs, LC for the first year). 5. LAI series with different increasing rate	Nine unregulated catchments in Yellow River Basin, China	Each percent increase of LAI results in: a) average change of around −0.2% for magnitude signatures, b) changes of -0.34 ± 0 c) wide ranges of +0.20%– + 0.99% for low flow days and −1.09% – −0.09% for high flow days, and; d) marginal change on flow dynamic signatures	This study

ence of greening on runoff processes, 10 runoff signatures are selected concerning magnitude, frequency, duration and dynamics in this study.

Because runoff simulated by a single model simulation may be biased and the related model uncertainty is difficult to quantify (Gudmundsson et al., 2012; Neuman, 2003), multimodel ensemble approaches have been applied to reduce the uncertainties for improved model performance (Ajami et al., 2006; Kaune et al., 2020; Neto et al., 2018; Velazquez et al., 2010; Yan et al., 2021). For example, Dong et al. (2021) developed a flow-based weighting strategy to recreate empirical flow frequency distributions. Meanwhile, Wan et al. (2021) suggested that the runoff performance can be improved by using less than the ensemble size of six, along with poor performance in terms of low flow. In addition, a time-varying-weight ensemble method generally provides better results than simple weighted methods (SWMs) under nonstationary conditions (Il Jeong & Kim, 2009). Therefore, it is reasonable to develop an ensemble method based on existing studies to focus on reducing uncertainties at multi-

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ple flow characteristics by considering both runoff magnitude and runoff process (especially for low flow). Based on this ensemble method, it can further investigate runoff change induced by greening.

In this context, the flow-based weighting strategy is advanced using a log-based weighted ensemble method (LWE) to improve low flow performance, since most of the selected runoff signatures largely depend on the daily flow duration curve (FDC) which is related to daily flow frequency. Moreover, improving the representations of vegetation dynamics in rainfall-runoff models is crucial for exploring greening impacts on runoff signatures (Ma et al., 2017). Therefore, five extensively used hydrological conceptual models are improved with an evapotranspiration model (Penman-Monteith-Leuning V2, PML-V2) to incorporate vegetation inputs (LAI and LC). Three ensemble methods, including the simple ensemble mean method (SEM), the flow-based weight method (abbreviated as the SWM, hereafter (Dong et al., 2021)) and LWE, are then applied based on the five improved models in this study.

The objectives of this study are to:

1. Reproduce runoff series with LWE to obtain better performance on both overall results and hydrological signatures;
2. Assess how much the greening impacted hydrological signatures in a selected study area with simple vegetation Modeling experiments; and
3. Estimate the sensitivity of hydrological signatures to greening.

In summary, this study differs from previous studies in the following aspects:

1. Considering not only runoff magnitude but also frequency, duration and dynamic signatures;
2. Improving hydrological models with PML-V2 ORCID to incorporate vegetation inputs; and
3. Using the ensemble methods to reduce potential simulation bias and to reproduce runoff signatures more accurately.

2. Methods

To explore the greening influence on runoff signatures, three major steps are conducted in this study and summarized in a flowchart (Figure 1). First, a robust approach to measure runoff time series and runoff signatures is chosen after cross-validation and performance evaluation. Second, modeling experiments of different vegetation inputs are designed to obtain different streamflow outputs. Finally, the greening influence is quantified by calculating the difference among runoff signatures simulated from different experiments. More specifically, this study quantified the following impacts: (a) LAI-induced runoff signature change, (b) LC-induced runoff signature change, (c) LAI and LC combined influences on runoff signature, and (d) the LAI and LC interaction influence on runoff signatures. Moreover, a sensitivity analysis of runoff signatures to LAI increase has also been accomplished by a series of different LAI inputs. All the details of the above methods are described in Sections 2.1–2.4.

2.1. Daily Runoff Simulation

Five individual conceptualized hydrological models incorporated with an evapotranspiration model (Penman-Monteith-Leuning V2, PML-V2) are applied to simulate runoff in this study. Then, SEM and SWM strategies are applied to the outputs of these five models to improve model performance. In addition, we further develop an LWE method based on SWM, aiming to improve the runoff performance on hydrological signatures, especially for low flows. Therefore, there are eight simulated runoff time series in total, obtained from (a) GR4J_PML, (b) HBV_PML, (c) HYMOD_PML, (d) SIMHYD_PML, (e) XAJ_PML, (f) SEM, (g) SWM, and (h) LWE.

2.1.1. Improved Hydrological Models With Better Representation of Evapotranspiration

The Penman-Monteith (PM) equation (Monteith, 1965) has been widely used for evapotranspiration (ET) estimates and many approaches have been developed to improve the model (Gan et al., 2018; Leuning et al., 2008; Mu et al., 2007; Pereira et al., 2015; Zhang et al., 2019). The PML (Penman-Monteith-Leuning) ET model is a typical type within the PM equation framework. First, surface conductance is used to account for the physical features in the PML ET model (Leuning et al., 2008). Then, the evaporation from the intercepted precipita-

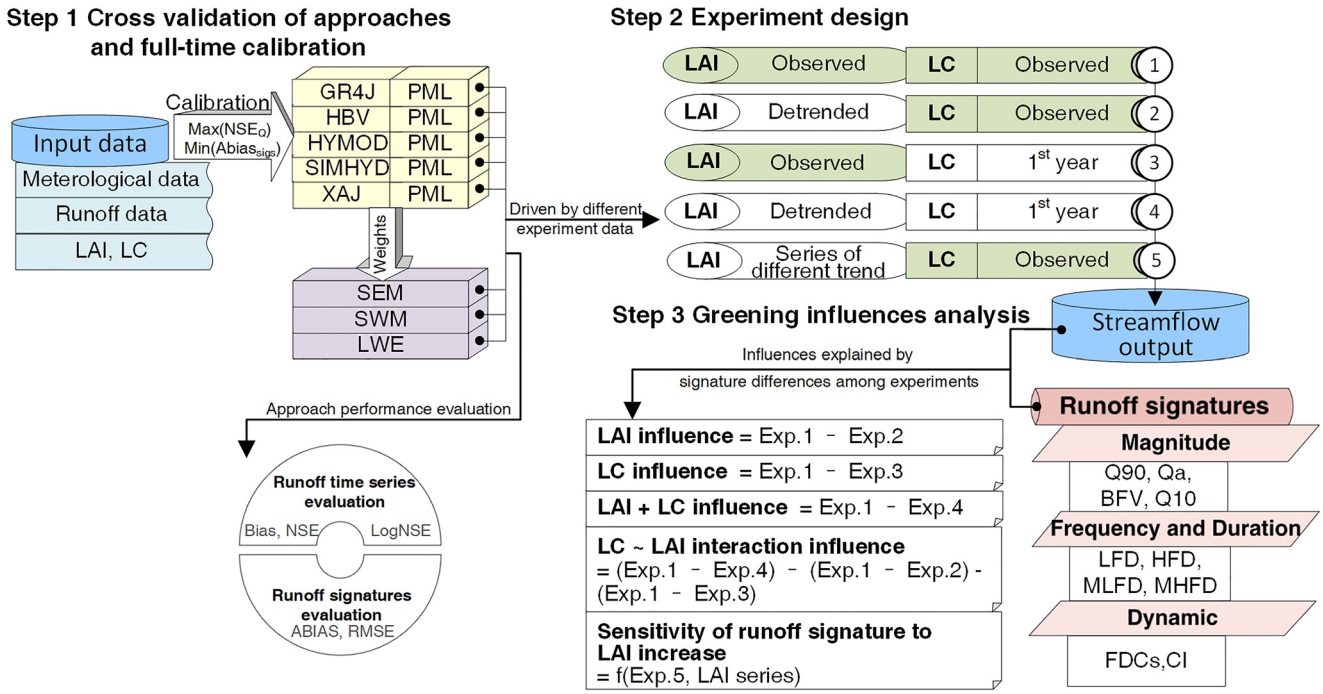


Figure 1. Framework for estimating vegetation greening influences. Exp.x: Signatures of the *x*th Experiment. Bias, Nash-Sutcliffe efficiency, LogNSE, absolute bias and root-mean-square error represent evaluation metrics. Qa, Q90, BFV, Q10, low flow day, high flow day, Max low flow duration, Max high flow duration, flow duration curves, and CI are 10 runoff signatures, and their details are summarized in Table 2.

tion is included as an important component of PML model in PML-V1 (Zhang et al., 2016). Furthermore, Gan et al. (2018) coupled ET with gross primary production (GPP) using a biophysical canopy conductance model (resulting in the PML-V2 ET model). Finally, Zhang et al. (2019) further improved PML-V2 by incorporating the vapor pressure deficit constraint to constrain GPP, canopy conductance, and transpiration. The inputs of the PML-V2 model include meteorological (precipitation, incoming shortwave and longwave radiation, vapor pressure, air temperature, wind speed, CO_2 concentration) and remote sensing (LAI, albedo and surface emissivity) forcing data. The outputs of the model include evaporation of precipitation intercepted by vegetation (E_i), evaporation from the soil (E_s), transpiration from the plant canopy (E_t), and GPP. The 11 free parameters are listed in Figure S1 in Supporting Information S1. PML-V2 performs well in estimating ET across the globe (Zhang et al., 2019) and across certain hot-pot regions, including the Tibetan Plateau (Ma & Zhang, 2022) and the Yellow River Basin (Li, Zhang, et al., 2020; Yin et al., 2020).

The ET modules of all five conceptualized hydrological models are replaced by the PML-V2 model in this study. The five hydrological models include GR4J (Perrin et al., 2003), HBV (Bergstrom & Lindstrom, 2015; Bergström, 1995; Seibert, 1997), HYMOD (Boyle, 2001; Wagener et al., 2001), SIMHYD (Chiew et al., 2002) and XAJ (Zhao, 1980, 1992). All five models contain a soil storage module to regulate soil water. To better replace ET modules in hydrological models with the PML-V2 model, we first divided the PML-V2 output into two parts: E_i and E_{st} (the sum of soil evaporation and transportation, $E_{st} = E_s + E_t$). Here, E_i is constrained according to precipitation and canopy storage. E_{st} is constrained by soil storage. Finally, the actual ET equals the sum of constrained E_i and E_{st}. The structures of the PML-V2 model and each improved model are shown in Figures S1 to S6 in Supporting Information S1.

2.1.2. Simple Ensemble Mean (SEM) Method and Simple Weighted Mean (SWM) Method

Numerous studies found that the SEM of a group of models could give the best runoff performance compared to an individual model (Il Jeong & Kim, 2009; Velazquez et al., 2010; Wan et al., 2021). This is especially true for some evaluation metrics because the model combination can compensate for the weakness of hydrological model structures. However, the overall effectiveness of all aspects of runoff characteristics has not been assessed. Since most of the runoff signatures selected in this study (see Section 2.2) can be derived from the FDC, incorporating

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another ensemble method using the FDC based weighted method may be better for describing runoff signatures. Dong et al. (2021) proposed the SWM to robustly assess the impacts of climate change on runoff signatures combined with six global circulation models. This method calculates the weight of each model by quantifying the discrepancies between the FDCs from observed flows and those from simulated flows. The larger the discrepancy is, the smaller the coefficients, and vice versa. In this way, weighted coefficients are assigned to each model to calculate the final weighted mean results.

2.1.3. Improvement of the Simple Weighted Mean Method

Since the SWM method (Dong et al., 2021) improves the simulated FDC by using the entire discrepancy between the observed and simulated FDCs, the skill can be further improved by dividing them into several parts. Therefore, we further developed the LWE method based on SWM by dividing daily runoff into two parts according to the FDC.

First, the FDC can be divided into two parts through a threshold of nonexceedance probability equals to 0.5, that is, the high flow part (nonexceedance probability >0.5) and the low flow part (non-exceedance probability ≤0.5) of the FDC for observed runoff and simulated runoff from the five improved models. For the FDC high flow part, the discrepancy of each model was calculated with absolute values. For the low flow part, the log transformation was used to enlarge the discrepancy of low flows, thus allocating more weight to low flows. Finally, based on the discrepancies, we developed a weight calculation scheme for each model as follows:

$$Q_{\mathrm{LWE}} = \begin{cases} \sum_{i=1}^{M} W_{i,\mathrm{low}} Q_{\mathrm{sim},i}, \ (Q < Q_{p=0.5}) \\ \sum_{i=1}^{M} W_{i,\mathrm{high}} Q_{\mathrm{sim},i}, \ (Q \geq Q_{p=0.5}) \end{cases} \tag{1}$$

$$\begin{cases} W_{i,\mathrm{low}} = \dfrac{DC_i}{\sum_{i=1}^{M} DC_i}; \ DC_i = \int_0^{0.5} |D| \, \mathrm{d}p = \int_0^{0.5} |\log(Q_{\mathrm{sim},i}) - \log(Q_{\mathrm{obs}})| \, \mathrm{d}p \\ W_{i,\mathrm{high}} = \dfrac{DC_i}{\sum_{i=1}^{M} DC_i}; DC_i = \int_{0.5}^{1} |D| \, \mathrm{d}p = \int_{0.5}^{1} |Q_{\mathrm{sim},i} - Q_{\mathrm{obs}}| \, \mathrm{d}p \end{cases} \tag{2}$$

where Q_{LWE} represents the simulated runoff series of the LWE, and $Q_{\mathrm{sim},i}$ represents the runoff series of the ith model among the M models. $W_{i,\mathrm{low}}$ is the weighted coefficient for FDC low flow whose nonexceedance probability <0.5 for the ith model, and vice versa. DC_i represents the sum of absolute discrepancy (D) for the ith model. Using LWE can improve low flow estimates compared to the original method (SWM). Figure 2 shows an example of FDC estimates obtained from eight approaches, which demonstrates that LWE is the most robust. Furthermore, FDC estimates for all nine catchments are summarized in Figure S7 in Supporting Information S1.

2.1.4. Performance of Eight Approaches

To obtain the eight outputs, the five improved models are first calibrated, and the weights of each model for the three ensemble methods are then calculated based on the five models. The performances of the eight approaches are evaluated in different temporal periods.

First, the five improved models should be calibrated. Each improved model contains parameters for both the PML-V2 model and the relevant hydrological model. To reduce the uncertainties caused by too many parameters, the PML-V2 parameters are preset by comparing LC types with the Resource and Environment Science and Data Center (RESDC) and MODIS (Friedl & Sulla-Menashe, 2019). These preset parameters were obtained by calibration for 10 LC types through 96 global flux towers by Zhang et al. (2019). The parameters to calibrate are listed in Figures S2–S6 in Supporting Information S1.

The objective function used for model calibration focused on both runoff time series simulation and runoff signature estimation. Thus, the Nash-Sutcliffe efficiency (NSE) of daily runoff (Nash & Sutcliffe, 1970) and the absolute bias (ABIAS) of runoff signatures (see Section 2.2 for details) are applied to optimize both runoff time series and runoff signatures, respectively. The ranges of NSE and ABIAS are $(-\infty, 1]$ and $[0, +\infty)$, respectively. The smaller the values of (1−NSE) and ABIAS, the better the performance. Therefore, a multiobjective function is set to minimize both NSE and ABIAS as follows:

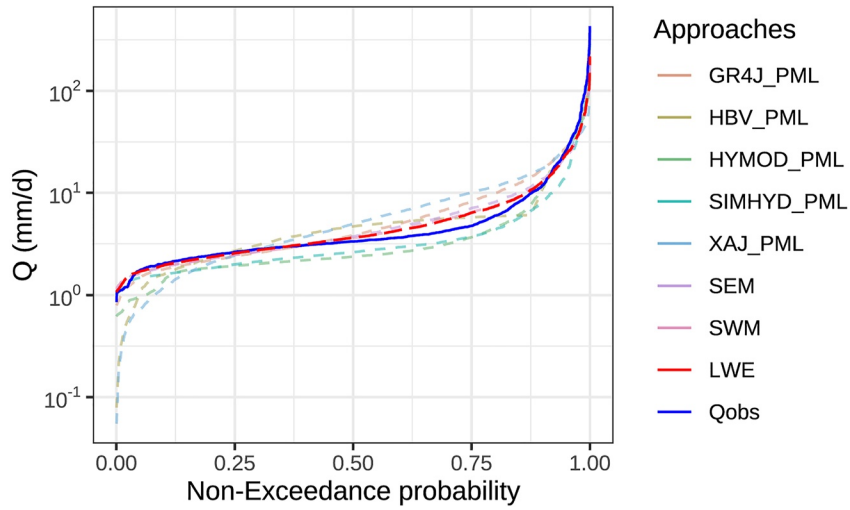


Figure 2. An example (Qinduzhen, Catchment #4 in Section 3.1) showing the observed and estimated flow distribution curve (FDC) to demonstrate the advantage of log-based weighted ensemble (LWE). Simple ensemble mean method allocates the same weights for the five hydrological models. Simple weighted method allocates the weights by the discrepancy between each hydrological model simulation and the observed FDC. LWE contains two sets of weights of five hydrological models by dividing them into two parts. The LWE method minimizes the discrepancy between the approach-simulated FDC and the observed FDC.

$$F_1 = 1 - \mathrm{NSE} + \sum_{k=1}^{n} \mathrm{ABIAS}_k = \frac{\sum_{j=1}^{m}(Q_{\mathrm{obs},j} - Q_{\mathrm{sim},j})^2}{\sum_{j=1}^{m}\left(Q_{\mathrm{obs},j} - \overline{Q_{\mathrm{obs}}}\right)^2} + \sum_{k=1}^{n}\left|\frac{(\mathrm{Sig}_{\mathrm{sim},k} - \mathrm{Sig}_{\mathrm{obs},k})}{\mathrm{Sig}_{\mathrm{obs},k}}\right| \tag{3}$$

where $Q_{\mathrm{obs},j}$ and $Q_{\mathrm{sim},j}$ represent the jth observed daily runoff (mm/d) and the jth simulated daily runoff (mm/d), respectively, and m represents the total number of records. $\mathrm{Sig}_{\mathrm{obs},k}$ and $\mathrm{Sig}_{\mathrm{sim},k}$ represent the kth observed runoff signatures (units following the unit of each signature) and the kth simulated runoff signatures (units following the unit of each signature), respectively, and n represents the total number of signatures. Note that the NA values of ABIAS caused by zero $\mathrm{Sig}_{\mathrm{obs},k}$ are removed during calibration.

Each model is warm-up from the first 2 years before the observed runoff time series starts. Two-thirds of the daily runoff data are randomly selected for calibration, and the remaining data are used for validation. The data for corresponding dates are also used for model inputs. Then, the weighted factors of LWE and SWM are calculated according to Sections 2.1.3 and 2.1.4. Based on the outputs of five improved models, SEM, SWM, and LWE, the eight approaches are evaluated using Bias, LogNSE, and NSE for the calibration and validation periods.

After obtaining a satisfactory result in the validation period (Section 4.1.1), the five models are then recalibrated using the full period of data to make full use of the available data and the three ensemble methods are subsequently applied. The eight full-time outputs are also evaluated using Equations 4–6:

$$\mathrm{Bias} = \frac{\sum_{j=1}^{m}(Q_{\mathrm{sim},j} - Q_{\mathrm{obs},j})}{\sum_{j=1}^{m}(Q_{\mathrm{obs},j})} \tag{4}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{j=1}^{m}(Q_{\mathrm{obs},j} - Q_{\mathrm{sim},j})^2}{\sum_{j=1}^{m}\left(Q_{\mathrm{obs},j} - \overline{Q_{\mathrm{obs}}}\right)^2} \tag{5}$$

$$\mathrm{LogNSE} = 1 - \frac{\sum_{j=1}^{m}(\log(Q_{\mathrm{obs},j}) - \log(Q_{\mathrm{sim},j}))^2}{\sum_{j=1}^{m}\left(\log(Q_{\mathrm{obs},j}) - \overline{\log(Q_{\mathrm{obs}})}\right)^2} \tag{6}$$

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where the variables are the same as those in Equation 3, but for the evaluation periods.

Next, the performance of runoff signatures (which will be introduced in Section 2.2) for each approach and each signature can be estimated using the ABIAS and root-mean-square error (RMSE).

$$\mathrm{ABIAS}_{i,k} = \frac{\sum_{l=1}^{N} |\mathrm{Sig}_{\mathrm{sim},l} - \mathrm{Sig}_{\mathrm{obs},l}|}{\sum_{l=1}^{N} \mathrm{Sig}_{\mathrm{obs},l}} \tag{7}$$

$$\mathrm{RMSE}_{i,k} = \sqrt{\frac{\sum_{l=1}^{N} (\mathrm{Sig}_{\mathrm{sim},l} - \mathrm{Sig}_{\mathrm{obs},l})^2}{N}} \tag{8}$$

where i and k represent the ith approach and the kth runoff signature, N represents N catchments, $\mathrm{Sig}_{\mathrm{sim},l}$ represents simulated signatures for the ith approach and the kth runoff signature at the lth catchment, and $\mathrm{Sig}_{\mathrm{obs},l}$ represents observed signatures for the kth runoff signature at the lth catchment. It should be noted that ABIAS is excluded in both the calibration and evaluation of some catchments, where observed signatures are zero and Equation 7 becomes invalid. Thus, we only use those catchments with valid ABIAS values. In addition, using RMSE can avoid the above issues and give a fairer evaluation of modeling results.

Finally, since the range of RMSE among 10 signatures are different from each other, to better compare the signature performance of eight approaches in one radar plot, a normalized method should be applied. The RMSE values are then grouped according to signatures, with the corresponding RMSE of the eight approaches for each group. Then, a normalized method is applied to compare the signature performance of the eight approaches within the group:

$$\text{Normalized RMSE}_{i,k} = \frac{(\mathrm{RMSE}_{i,k} - \min(\mathrm{RMSE}_k))}{(\max(\mathrm{RMSE}_k) - \min(\mathrm{RMSE}_k))} \tag{9}$$

where $\text{Normalized RMSE}_{i,k}$ represents the normalized RMSE for the ith approach and the kth runoff signature. $\min(\mathrm{RMSE}_k)$ and $\max(\mathrm{RMSE}_k)$ represent the minimum and maximum RMSE of the kth runoff signature across the M approaches, respectively. Equations 7–9 indicate that good performance for signatures should provide a smaller ABIAS and normalized RMSE (both approaching zero).

2.2. Runoff Signatures

A natural streamflow regime can be divided into five components, including magnitude, frequency, duration, timing, and rate of change (Poff et al., 1997; Richter et al., 1996). These components can be expressed as runoff signatures to characterize flows explicitly and have been widely considered (McMillan, 2021; Zhang, Chiew, et al., 2018). For example, the flow magnitude signatures characterize the flow amount of a given frequency within a temporal period. The flow frequency signatures refer to how often a given magnitude reoccurs. Flow duration and timing signatures refer to the regularity with which they occur for a given magnitude. The rate-of-change signatures give the dynamic information of how quickly flow changes. Therefore, using runoff signatures covering these components can quantify hydrological changes.

The choice of hydrological signatures should be considered comprehensively (McMillan, 2021). Here, we selected four types of signatures in this study, including magnitude signatures, frequency signatures, duration signatures and flow dynamic signatures. LFD and HFD in Table 2 are derived on multiyear average scale, which is a transformation of flow frequency signatures using the ratio between low/high flow days and total days. The runoff signatures examined are listed in Table 2.

2.3. Modeling Experiments for Quantifying the Impact of Greening on Runoff Signatures

Different modeling experiments are carried out to drive five models with calibrated parameters, and three ensemble approaches with fixed weights that have been calculated in the calibration stage. Then, the outputs of eight approaches are further used to estimate the influence of greening on runoff signatures. LAI and LC are usually regarded as two important vegetation-related inputs. Therefore, four modeling experiments are conducted:

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Table 2
Runoff Signatures Used in This Study

Type	ID	Signature	Long name	Units	Focused aspects	Details	Reference
Magnitude	1	Q90	Runoff in 90th quantile of FDC	mm/day	High flow	90th flow percentile of streamflow	
	2	Qa	Annual runoff	mm/year	Mean annual flow	Mean annual flow	
	3	BFV	Base flow volume	mm/year	Base flow	Annual base flow is separated from total streamflow by UKIH methods (also known as Smoothed Minima Method)	(UK Institute of hydrology, 1980)
	4	Q10	Runoff in 10th quantile of FDC	mm/day	Low flow	10th Flow percentile of streamflow	
Frequency and duration	5	LFD	Low flow days	Days (multiyear average)	Days of potential water shortage	Multiyear average of days whose runoff below 0.2*mean daily runoff	(Addor et al., 2018; McMillan, 2021; Olden & Poff, 2003)
	6	HFD	High flow days	Days (multiyear average 1)	Days of potential flood	Multiyear average of days whose runoff above 0.9*median daily runoff	(Addor et al., 2018; McMillan, 2021; Olden & Poff, 2003)
	7	MLFD	Max low flow duration	Days (multiyear max)	The risk of extreme water shortage	Maximum number of consecutive LFD over the period	
	8	MHFD	Max high flow duration	Days (multiyear max)	The risk of continuous flood	Maximum number of consecutive HFD over the period	
Dynamic	9	FDCs	Slope of flow duration curve (FDC)	–	Rate of FDC slope change	Slope of a range from 0.33 to 0.66 for FDC	(McMillan et al., 2017; Yadav et al., 2007)
	10	CI	Index describing the shape of FDC	–	Rate of FDC shape change	Concavity index (Q90-Q1)/(Q99-Q1)	(Zhang, Chiew, et al., 2018)

1. Actual condition (observed LAI and LC inputs),
2. LAI-detrended condition (observed LC inputs, detrended LAI),
3. LC-detrended condition (observed LAI inputs, LC held constant at its first-year value); and
4. LAI-LC-detrended condition (detrended LAI inputs, LC held constant at its first-year value).

Apart from LAI and LC data, all of the above four experiments share the same inputs for the whole modeling period (from 2002 to 2017). Moreover, the model parameters are the same as the calibrated parameters for each model despite different modeling experiments. For each experiment, five models and three ensemble methods are applied to generate eight kinds of outputs in total. The influence of greening on runoff signatures is then quantified by calculating the signature difference between actual vegetation experiment and detrended vegetation experiments (Figure 1). For example, LAI-induced runoff signature change can be explained by the difference between the runoff signatures of Experiment 1 and Experiment 2 (Exp. 1–Exp. 2). LC-induced runoff signature change equals Exp. 1–Exp. 3. Their combined influence equals Exp. 1–Exp. 4, and their interacted influences can be explained by the combined influence minus the separated ones.

Note that for frequency and duration signatures, the low (or high) flow threshold of the four experiments is set as the same value of low/high flow of Exp. 4 for an easier comparison and interpretation. For instance, the low flow threshold is set as 0.2 × mean daily runoff in Exp. 4 (LC-LAI detrended) and the high flow threshold is set as 0.9 × median daily runoff in Exp. 4 (LC-LAI detrended). The daily runoff for calculating runoff signatures spans from 2003 to 2017 due to LAI data deficit in 2002 (LAI data starts from *04 July* 2002, and the missing values before *04 July* 2022 were filled with mean annual LAI for model simulation).

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2.4. Sensitivities of the Runoff Signatures to Increasing LAI

LAI series of different increasing trends are designed in Exp. 5 (Figure 1) to further analyze the possible runoff signature sensitivity under different LAI increasing rates. For each catchment, the inputs of Exp. 5 remain dynamic except for LAI. The LAI series are set as different increasing rates of 0–3 times of the actual rate with steps of 0.01 (Figure S8 in Supporting Information S1). This means, that 301 groups of LAI series with rates of 0–3 times of the actual rate are generated at each catchment. The detrended LAI (LAI series in Exp. 2 and Exp. 4) and observed LAI series (LAI series in Exp. 1 and Exp. 3) are the same as the LAI series with zero and one times of rate, respectively. Then, the corresponding streamflow output of each series will be applied to generate runoff signatures in Exp. 5. Finally, scatterplots of runoff signatures against mean LAI are easy to draw, which show a strong linear relationship between the two variables for each basin (Figure S9 in Supporting Information S1). Therefore, the absolute sensitivity of each catchment can be easily estimated via the slope of the linear regression model between runoff signatures and the means of the LAI series (Equation 10). Consequently, the relative sensitivity can be estimated via Equation 11.

$$\mathrm{Sig} = \mathrm{sensitivity}_{\mathrm{absolute}} * \mathrm{LAI} + \varepsilon \tag{10}$$

$$\mathrm{Sensitivity}_{\mathrm{relative}} = \frac{\mathrm{dSig}}{\mathrm{Sig}_0} \Big/ \frac{\mathrm{dLAI}}{\mathrm{LAI}_0} = \frac{\mathrm{dSig}}{\mathrm{dLAI}} * \frac{\mathrm{LAI}_0}{\mathrm{Sig}_0} = \mathrm{sensitivity}_{\mathrm{absolute}} * \frac{\mathrm{LAI}_0}{\mathrm{Sig}_0} \tag{11}$$

where Sig represents the runoff signatures simulated by the 301 LAI groups, $\mathrm{sensitivity}_{\mathrm{absolute}}$ and $\mathrm{sensitivity}_{\mathrm{relative}}$ represents sensitivity of runoff signature to LAI in absolute and relative form, respectively, and ε is the interception of the linear regression model. d stands for the differential symbol. LAI_0 and Sig_0 represent detrended LAI and runoff signatures of detrended LAI, respectively.

3. Study Area and Data

3.1. Study Area

The study area is selected within the Wei River Basin (WRB), the largest tributary of the Yellow River in China, which provides most water resources for the Guanzhong Plain, in which a large amount of water is needed for both agricultural and municipal activities (Zhao et al., 2013). It has been widely reported that the vegetation in the WRB has been greening in recent decades (Gao et al., 2019; Wang, Sun, et al., 2017). Therefore, quantifying the influence of greening on runoff is extremely important for the water resource management of this basin. Nine catchments within the WRB are selected by the following criteria to ensure they are largely unregulated: (a) few large water conservancy facilities determined by the information retrieved from GRanD (Lehner et al., 2011); (b) the storage capacity of all reservoirs is less than 10% of the catchment mean annual runoff. Catchments #1–3 are the upper catchments of the WRB. It should also be noted that catchments #1 and #2 are nested within catchment #3, and catchment #5 is nested within catchment #6. To better understand vegetation change in the study area, the LAI and LC data are presented in Figures 3b and 3c. Figure 3b shows the LC type changes from crop to forest or grass during from 2000 to 2018. The black line in Figure 3c shows that the LAI in the study area increases by approximately 0.02 per year.

Table 3 gives more information about the nine catchments, which are located in arid and semiarid regions, with a high aridity index (AI), low annual precipitation and runoff depths.

3.2. Data

Table 4 shows details of the data used in this study. The climate forcing variables used for driving the model are from the Climate Meteorological Forcing Data set (CMFD), which includes near-surface air temperature, specific humidity, air pressure, 10-m wind speed, precipitation, and downward shortwave and longwave radiation across China (He et al., 2020). The CMFD is accurate meteorological forcing data in China and has therefore been widely used for ET and runoff simulations in China (He et al., 2020, 2021; Ma, Szilagyi, et al., 2019; Ren et al., 2018; Yang et al., 2017). Daily maximum temperature and minimum temperature are generated from 3-hr CMFD data, while other elements are aggregated to the daily scale.

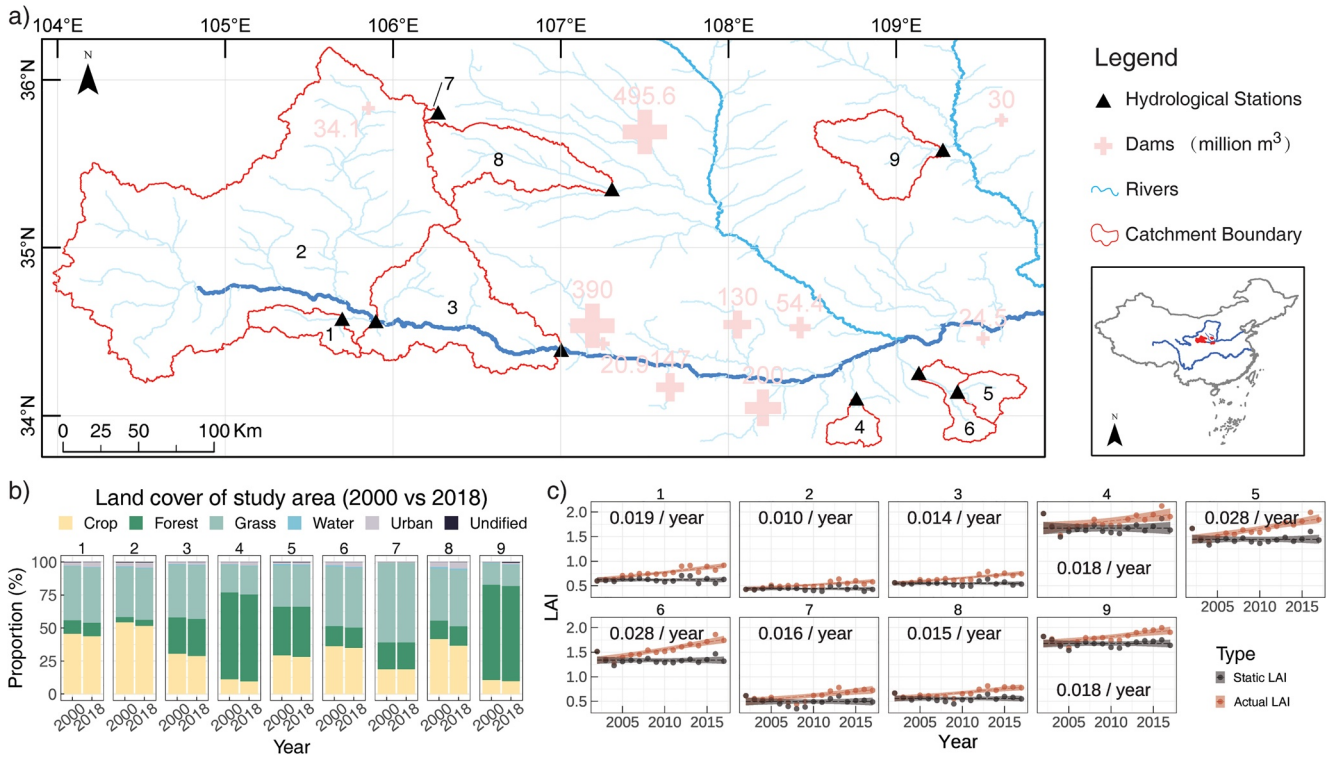


Figure 3. Location of the study area. (a) Locations of the study area, (b) land cover changes for the study area, (c) actual annual leaf area index (LAI) and detrended LAI.

The surface LAI, albedo and emissivity data are all from Moderate Resolution Imaging Spectroradiometer (MODIS V6). The raw LAI data are smoothed by a Whittaker smoother-based method developed to generate a more reasonable actual LAI series (Kong et al., 2019; Zhang et al., 2019). The smoothed LAI, raw albedo and raw emissivity are interpolated by linear and historical average interpolation methods, which are also used in the study of Zhang et al. (2019). LAI is later detrended (Bai et al., 2020), which means that the seasonal variation is maintained, with only the annual trend removed.

The LC data are obtained from the Resource and Environment Science and Data Center, China. The LC is divided into six types: crop, forest, grass, water, urban and undefined area. The LC data proportion of each type for each

Table 3
Summary of Catchments Used for This Study

ID	Name	Lon (°)	Lat (°)	AREA (km^2)	Elevation (m)	Runoff (mm/yr)	Precipitation (mm/yr)	PET (mm/yr)	AI (−)
1	Tianshui	105.70	34.58	1020	1,713	59	539	874	1.62
2	Beidao	105.90	34.56	25,046	2,012	23	476	833	1.75
3	Linjiacun	107.00	34.39	30,718	1,934	38	523	806	1.54
4	Qinduzhen	108.76	34.10	572	1,490	448	747	749	1.00
5	Luolicun	109.37	34.14	745	1,350	307	721	1,016	1.41
6	Maduwang	109.13	34.25	1,623	1,159	314	709	1,018	1.44
7	Longde	106.27	35.81	41	2,374	58	437	832	1.90
8	Jingchuan	107.31	35.35	3,145	1,681	45	465	905	1.95
9	Huangling	109.28	35.58	2,281	1,302	55	645	908	1.41

Note. PET represents potential ET calculated by the reference potential evapotranspiration method (Allen et al., 1998); AI, aridity index, is calculated by the mean annual PET divided by the mean annual precipitation.

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Table 4
Data Summary Used in This Study

Data sources	Short name	Long name	Temporal resolution	Spatial resolution	Temporal coverage	References
MODIS/006/MCD15A3H	LAI	Leaf Area Index	4-day	500 m	04 July 2004 to present	(Myneni et al., 2015)
MODIS/006/MCD43A3	Albedo	Albedo	Daily	500 m	18 February 2000 to present	(Schaaf & Wang, 2015)
MODIS/006/MOD11A2	Emiss	Emissivity	8-day	1 km	05 March 2000 to present	(Wan et al., 2015)
CMFD (The Climate Meteorological Forcing Data set)	T_{max}	Max temperature	3 hr	0.1°	1979–2018	(He et al., 2020; Yang et al., 2010)
	T_{min}	Min temperature				
	Pa	Air Pressure				
	U	Wind speed at 10-m height				
	q	Specific humidity				
	Prcp	Precipitation				
	Rln	Incoming longwave radiation				
	Rs	Incoming shortwave solar radiation				
Data Center for Resources and Environmental Sciences, Chinese Academy of Sciences (RESDC)	Landcover	Resdc Landcover	Around every 5 years	30 m	1970s, 1980s, 1995–2015 every 5 years, 2018	–
NOAA globally averaged monthly mean data	CO_2	Carbon dioxide density	Monthly	Global	1958 to present	–
Yearbook of Yellow River Basin	Runoff	Runoff	Daily	Gauges	Variation among the hydrological gauges (2002–2012)	–

Note. The temporal coverage shown in this table represents the temporal range of data sets. The data temporal coverage used for this study is described in Section 2.

catchment in 2000 and 2018 is shown in Figure 2b. Furthermore, the LC proportion is updated approximately every 5 years for model inputs according to the raw LC data. The global monthly CO_2 data is also used to drive PML-V2.

The daily observed runoff data of the nine hydrological stations are provided by the Yearbook of Yellow River Basin, covering the period 2002 to 2012.

4. Results

4.1. Runoff Simulations

4.1.1. Overall Performance

The performance of calibration using two-thirds of the data and validation using the remaining data is shown in Figure S10 in Supporting Information S1. Figure 4 summarizes the performance of eight simulated runoff time series obtained from five models and three ensemble methods against full-time calibration results. That is, the eight approaches applied to nine catchments are assessed using three metrics (Figures 4a–4c, Table S1 in Supporting Information S1). Figure 4a shows that LWE gives small Bias values (-0.038 ± 0.055) in the nine catchments and outperforms five approaches with a relatively narrow spread. The LWE performs best for estimating low flows, as evidenced by the highest LogNSE value (0.441 ± 0.077) across the nine catchments (Figure 4b). The LWE also yields the best NSE values in the nine catchments (0.660 ± 0.007). The spread of the above values of LWE in these catchments is particularly small compared to other methods, suggesting that it is a stable approach in depicting low flow in different catchments. In summary, the three ensemble approaches perform better than the individual hydrological models, and LWE performs well in not only overall performance

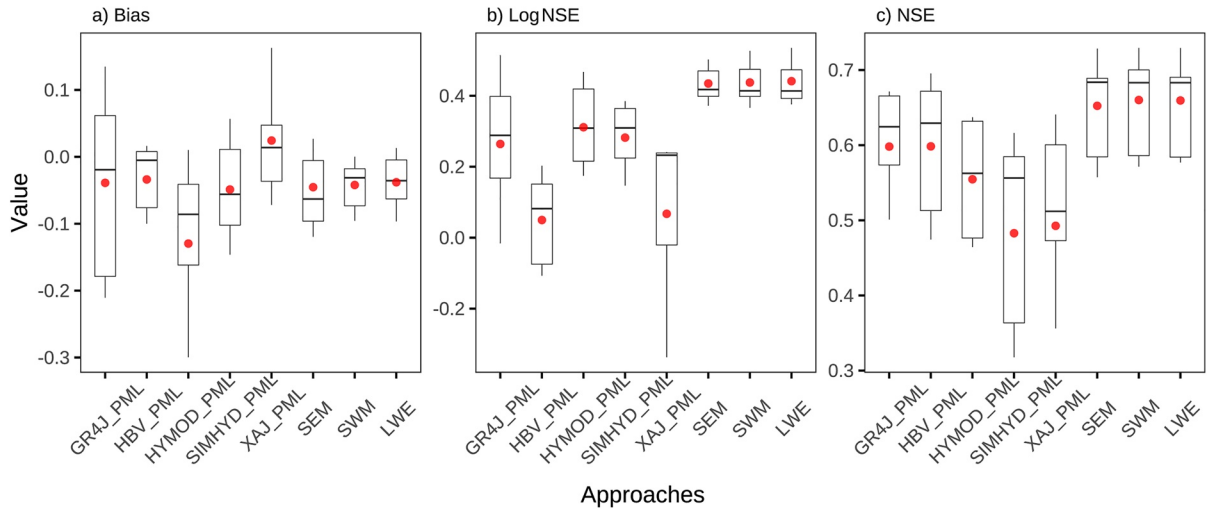


Figure 4. Performance of eight approaches from five individual models and three ensemble methods. The mean metric value is shown as red dots. The three horizontal lines in each box represent the values of the 25th, 50th and 75th percentiles. The whiskers range from the 10th to 90th percentiles.

but also in representing the low flow, a hydrologic component that is particularly difficult to model when also concerning high flow (Oudin et al., 2006; Pushpalatha et al., 2012).

4.1.2. Performance on Runoff Signatures

In addition to the overall performance in runoff, the performance of the eight approaches to reproduce 10 runoff signatures is illustrated in Figure 5 and Table S2 in Supporting Information S1. Figure 5a implies that LWE performs well for ABIAS of the 10 signatures (0.15 ± 0.08). The value does not increase much compared to the lowest ABIAS (HYMOD_PML, 0.10 ± 0.05). Regarding normalized RMSE (Figure 5b), LWE has the lowest mean normalized RMSE values across 10 signatures (0.13 ± 0.15) among the eight approaches. Moreover, since there is no approach performing best for simultaneously simulating 10 runoff signatures, for each approach, the number of its exceeded approaches is summarized using normalized RMSE. LWE outperforms 5.6 models on average, which is much superior to other approaches (see the last column of normalized RMSE in Table S2 in Supporting Information S1). Specifically, the normalized RMSE of LFD for LWE (0.28) outperforms the other two ensemble approaches (0.32 and 0.30 for SWM and SEM, respectively).

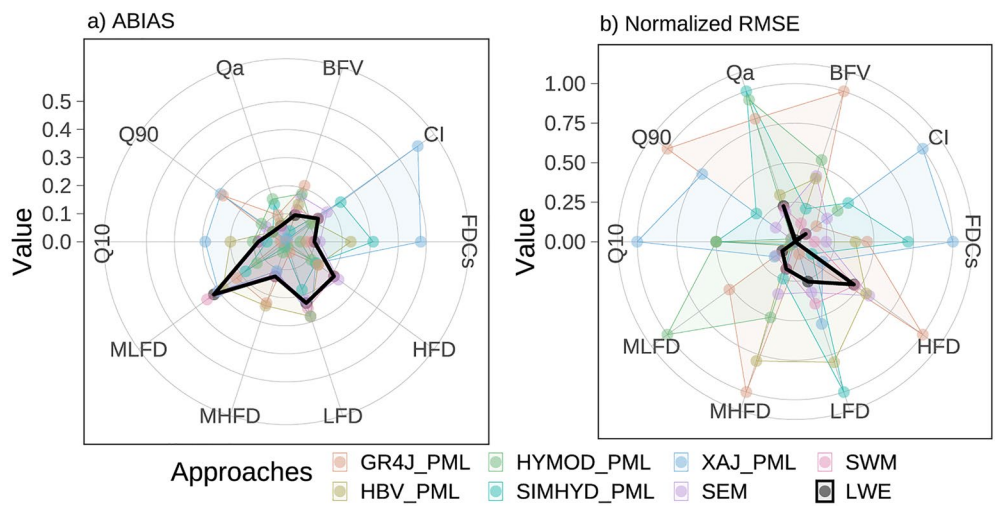


Figure 5. Performance of runoff signatures for eight approaches.

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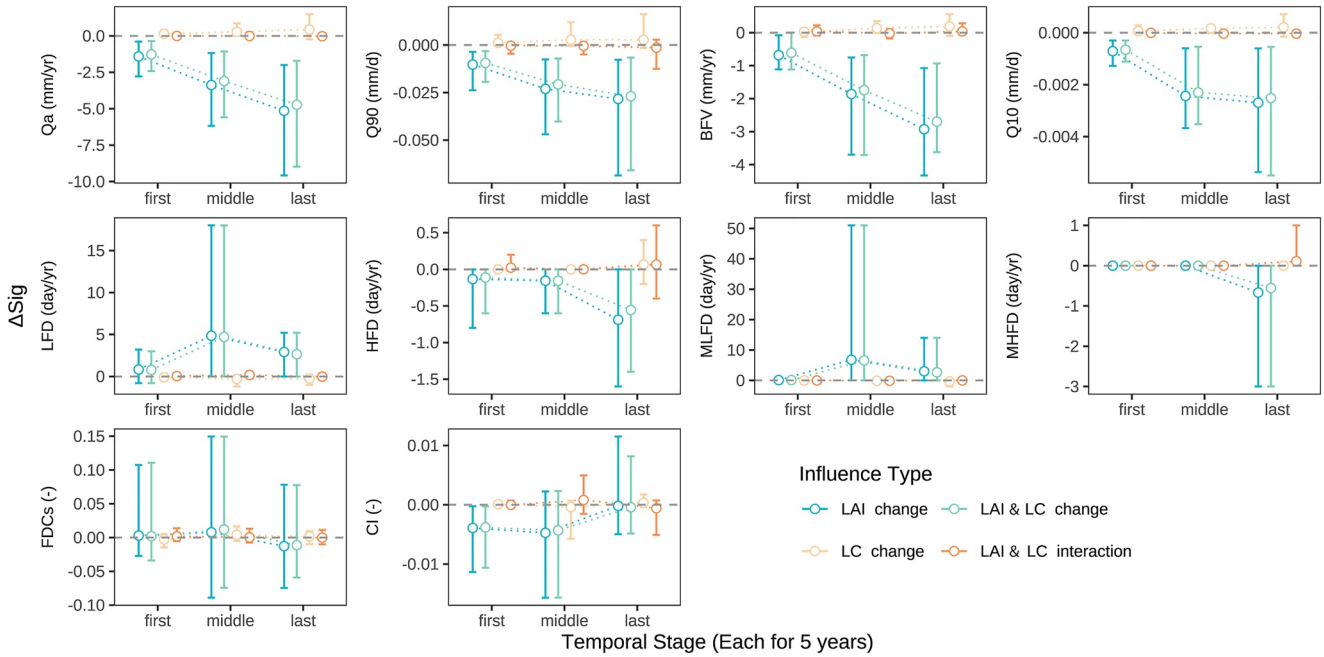


Figure 6. Greening-induced absolute runoff signature changes (ΔSig) of log-based weighted ensemble. The upper, middle and lower panels depict the magnitude, frequency/duration and dynamic signatures, respectively. The upper and lower boundaries of the error bars represent the maximum and minimum signature changes in the nine catchments, respectively. The circle represents the mean change in nine catchments.

In general, LWE is regarded as a robust approach in simulating various runoff signatures, and this method outperforms the other two ensemble methods, especially on low flow runoff signatures. LWE performs best not only in the overall time series evaluation (Figure 4), but also in the runoff signature simulation (Figure 5). For this reason, we mainly used LWE for subsequent runoff signature change analysis.

4.2. Temporal Analysis of Signature Change

The 15-year period during 2003–2017 is further equally divided into three 5-year periods: the first stage (2003–2007), middle stage (2008–2012) and last stage (2013–2017). Figure 6 summarizes greening (including LAI change/increasing, LC change, LAI and LC change, and the interaction influence of LAI and LC) induced runoff signature change (Figure 1) of the LWE representation of 10 hydrologic signatures on nine catchments for these three periods. Figure 6 suggests that the increasing LAI dominates the runoff signature change in these nine basins, while LC change results in little runoff signature change and adds little to LAI change-induced runoff signature change. The interaction influence of LAI and LC is also small, with a mean runoff signature change close to zero. Therefore, the following sections mainly focus on displaying the increasing LAI-induced runoff signature change.

In general, increasing LAI leads to a decline in the magnitude signature since changes in Qa, Q90, BFV, and Q10 are all negative (Figure 6). Meanwhile, the low flow frequency and duration signatures (LFD, MLFD) increased, while the high flow frequency and duration signatures (HFD, MHFD) did the opposite (Figure 6), suggesting an increasing risk for extremely dry events under greening. However, the increasing LAI shows a marginal influence on the flow dynamic signatures since both CI and FDCs vary little (Figure 6).

As shown in the upper panel, the four runoff magnitude signatures decrease more over time, and the mean ΔSig value of Qa (white circle) shows a similar pattern to that of BFV, indicating that the base flow volume contributes greatly to the mean annual runoff change in these nine catchments.

Furthermore, the signature change is not linear over time. The mean signature changes across the nine catchments are not strong for the first 5-year period (2003–2007) when compared with the last 10-year periods. Therefore, the following sections focus on the last 10-year period for LAI-induced runoff signature change analysis to explore the possible long-term hydrological consequences of vegetation greening.

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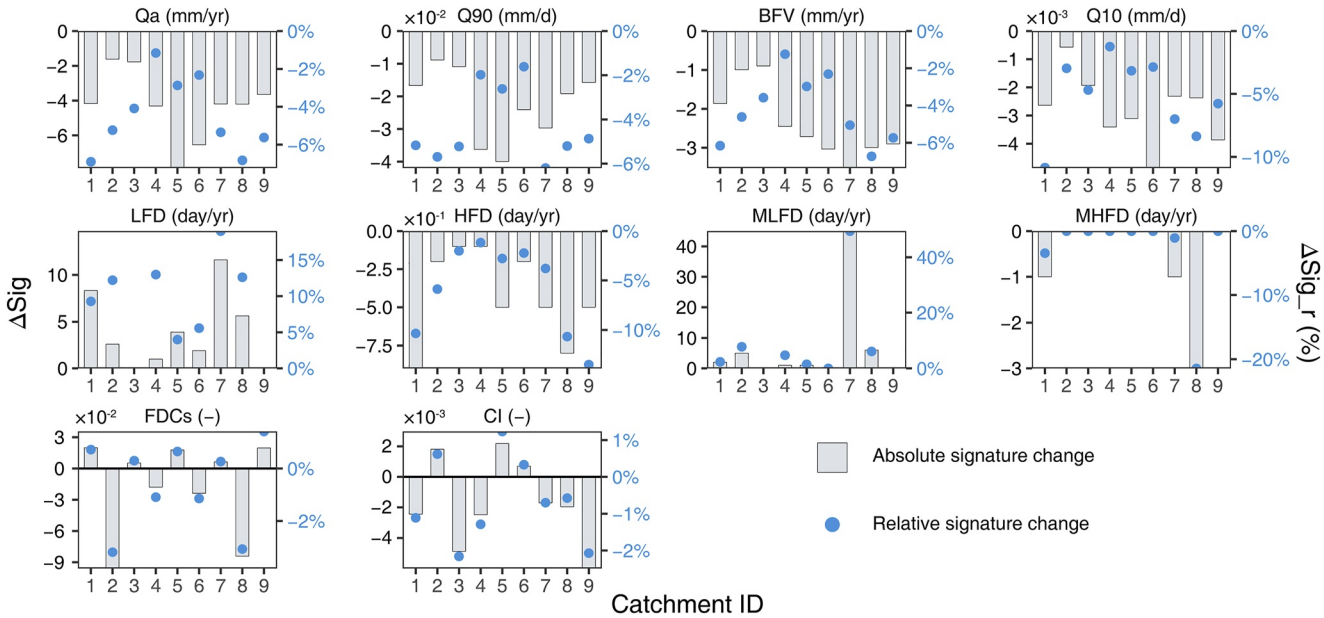


Figure 7. Vegetation-induced runoff signature change in absolute (ΔSig) and relative (ΔSig_r) form across nine selected catchments for the last 10 years.

4.3. Signature Changes for Last 10 Years

Figure 7 further shows the absolute signature change (ΔSig) and relative signature change (ΔSig_r) of LWE in each catchment for the last 10 years. The ΔSig_r is calculated by ΔSig divided by $Sig_{detrended}$. Although the magnitude signatures usually decrease, there are differences among catchments. Qa shares a similar pattern with BFV among catchments, and so do Q90 and Q10. The patterns for frequency, duration and dynamic signatures show no significant relationship with the catchment attribute. Same as explored in Figure 6, the change in frequency and duration signatures indicates an increasing risk for extremely dry events (e.g., LFD increases 11.6 days from 61.0 to 72.6 days at catchment #7) and reduction for extremely large floods (e.g., HFD reduces 0.5 days from 13.2 to 12.7 days at catchment #7). The flow dynamic signatures (CI, FDCs) vary little due to both positive and negative changes.

While the vegetation is greening, the relative changes in magnitude signatures over nine catchments are mostly approximately −4.5% (Table S3 in Supporting Information S1). For the frequency and duration signatures, relative changes in LFD range from 4.00% to 19.02% among all catchments, leading to an average increase of 10.81%. Other frequency and duration signatures also take a wide range: −13.51% to −1.16% for HFD, 0%–49.45% for MLFD, and −21.43%–0% for MHFD. Moreover, a high MLFD change relates to a high LFD change at catchment #7. Since the frequency and duration signatures are crucial to local environmental flow and irrigation, a considerable influence would be shown with the greatly changed frequency and duration signatures. Overall, the changes in the flow dynamic signatures are all relatively small, although the signs may be different in different catchments.

In summary, runoff magnitude signature changes vary greatly for absolute values, however, they share similar relative changes. Frequency and duration signatures change little for the absolute value, while relative changes may be to a large extent.

4.4. Sensitivities of Runoff Signature to LAI

The signature change rate with LAI increases is calculated as absolute (Figure 8) and relative (Figure 9) for each signature at each catchment for the last 10 years (the sixth to fifteenth year since LAI starts to change) based on Experiment 5.

As expected, all magnitude signatures decrease to some extent (Figure 8). In addition, the spatial distribution for each magnitude signature is even. More than half of the decrease in Qa (−21.00 ± 4.14 mm/yr) is due to a

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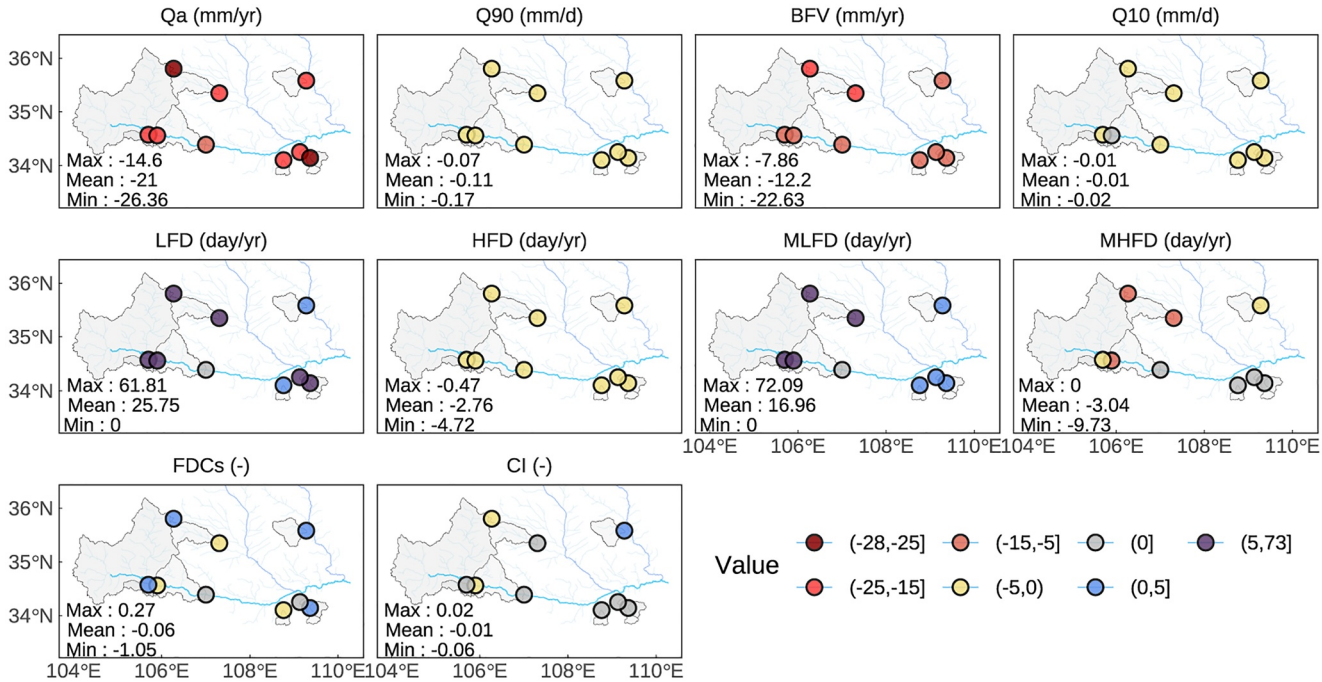


Figure 8. Signature change rate with increasing leaf area index.

decrease in BFV (−12.20 ± 4.70 mm/yr). The decreases in Q90 exceed those in Q10, suggesting that changes in high flow might be more apparent than those in low flow.

Frequency and duration signature change rates for low flow (LFD and MLFD) increase by 25.8 ± 25.7 days and 17.0 ± 24.2 days, respectively. At the same time, the frequency and duration signatures for high flow (HFD and MHFD) decreased by 2.8 ± 1.4 days and 3.0 ± 4.0 days, respectively. That is, more low flow days and fewer high flow days are obtained each year. As such, the low flow frequency and duration signatures are more sensitive than high flow ones to vegetation greening. Note that there were slight changes for certain catchments in frequency and duration signatures (zero signature change) because ET caused by LAI change is further constrained by soil water conservation, leading to a trivial frequency and duration signature change in some catchments. To summarize, the overall frequency and duration signature change indicates a reduction in water availability while LAI increases.

The dynamic signature change rate can be either positive or negative across different catchments, leading to mean values approaching zero for FDCs (−0.06 ± 0.40) and CI (−0.01 ± 0.02). Therefore, the flow dynamic change rate may be influenced by multiple basin properties, such as location and aridity, resulting in an unclear change in dynamic signatures.

Figure 9 gives the spatial relative runoff signature change rate of LWE for the last 10 years given that LAI increases by one percent. Basically, all magnitude signatures decrease no more than 0.5%, with mean change values of approximately −0.20%. Although LAI is an important input of the PML-V2 ET model, the calculated actual ET is later constrained by soil water storage in each hydrological model, and the model outputs are then weighted to obtain the LWE runoff series. Thus, one percentage increase in LAI would result in relative runoff magnitude signature changes less than 1% in most cases.

In terms of frequency and duration signatures, when LAI increased by 1%, the increases in LFD range from 0.17% to 0.99%, with a catchment mean of 0.56%. The MLFD increment ranges from 0% to 0.44%, with a mean value of 0.17%. The high flow frequency and duration signatures decrease less than the low flow ones' increase, with change values of −0.34 ± 0.30% and −0.08 ± 0.11% for HFD and MHFD, respectively. FDCs and CI show similar patterns compared with Figure 8.

In summary, the runoff signatures decrease by no more than 1% for every 1% increase in LAI. Compared to magnitude signatures, the LFD and HFD are more sensitive to LAI increases, a 1% increase in LAI could lead

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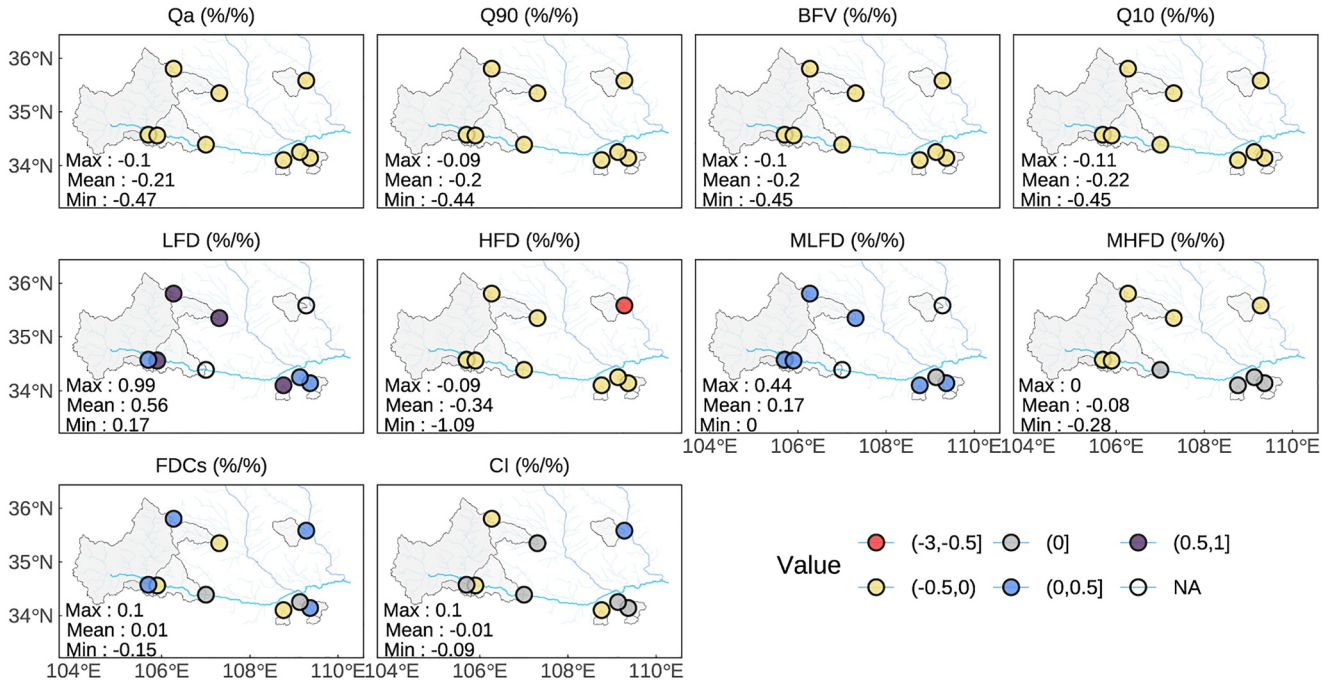


Figure 9. The signature change rate in percentage with leaf area index (LAI) increases with each percentage (Note that NA values means unable to calculate relative signature change due to zero signature value of the first LAI groups with zero rate.).

to a 0.56% increase in LFD and a 0.34% decrease in HFD on average, while there was only a 0.20% decrease for magnitude signatures. These results suggest that available water is reducing with greening impacts. For the dynamic signatures, changes in LAI did not lead to significant changes in FDCs and CI in most catchments.

5. Discussion

5.1. The Water Availability Risk of Greening on Runoff Signatures

This study isolated the influence of greening on runoff signatures based on model-simulated results among different modeling experiments. The LWE performs well in observed runoff signatures for both the calibration and validation periods (Figure S10 in Supporting Information S1), which proves that it is feasible to use the LWE for isolating changes in runoff signatures among the modeling experiments. The modeling experiments comparison approach used in this study has been widely used in isolating climate change factors (Mccarthy et al., 2010; Zhang et al., 2021), and is being increasingly used in catchment hydrological studies (Bai et al., 2020; Zhang, Wang, et al., 2020).

This study found that the water availability risk is growing with vegetation greening comprising runoff magnitude, high/low flow frequency, high/low flow duration and runoff dynamic. However, current studies mostly focused on runoff magnitude, and only a few on the runoff process. Figures 12c and 12d of Bai et al. (2020) indicate a significant difference (ranging from −10 to 0 mm/yr and −10% to 0%) in water yield when using actual and detrended inputs from 1982 to 2014 in the corresponding study area of this study (Figure 7). Their result agrees well with the Qa change shown in Figure 5 of this study. This study further indicates that the BFV accounts for 53 ± 16% of the annual runoff across nine catchments, thus the BFV reduction contributes approximately 55% of the annual runoff reduction on average among the nine basins with increasing LAI, which indicates a higher risk of base flow availability. It should be highlighted that the division of base flow volume agrees well with previous studies estimating the proportion of base flow to total runoff (35%–50%) in the Yellow River Basin (Cheng et al., 2019; Hu et al., 2021; Li & Zhang, 2017; Yanlin et al., 2008).

The decrease in Qa and BFV may result in water availability shortages, thus, low flow days may increase. This study proves this hypothesis with evidence of all positive low flow-related frequency and duration signature changes. Runoff frequency and duration signatures have been quantified not only for simulated runoff signature

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change analysis but also for sensitivity analysis to compare with the sensitivity of magnitude signatures. For every 1% increase in LAI, the frequency and duration signatures are more sensitive (e.g., 0.56% on average for LFD) and in a wider relative range (e.g., 0.17%–0.99% on average for LFD) than other signatures (basically no more than 0.2% for mean $|\Delta Sig|$). Given that the frequency and duration signature changes more rapidly and in a wider relative range than other signatures, a higher risk for available water shortages can result from greening. However, the dynamic signatures change little with greening.

In summary, unlike the literature, this study not only quantified the greening impact on runoff magnitude but also highlighted the greening impact on high/low frequency, high/low flow duration and runoff dynamics. The result further underlined the water availability risk of greening on runoff signatures. However, despite of LAI change, the water consumption is also determined by vegetation type and age, etc. As far as LC change is considered, recent literature gave similar results compared to the LC-induced runoff magnitude change in Figure 6 of this study. Zhang, Wang, et al. (2020) found that land use change contributes little to annual runoff change (0.2%) using the SWAT model combined with different scenarios from 1967 to 2017 in an Australian catchment. Although the catchments in Australia have different climate and geology conditions comparing to the catchments selected in this study, the two studies both have a similar conclusion, that is, little influence of LC on runoff signatures. LC change is complex, reflected by not only LAI but also more compound variables; thus, a more comprehensive analysis should be conducted.

5.2. How Did Other Greening Related Climate Variables Impact Runoff Signatures?

This study explores the runoff signature change influenced by LAI change, LC change, their combined change and their interaction. However, the greening caused by climate variables and other characteristics of catchments has not been investigated. A previous study showed that CO_2 fertilization effect at a global scale is the main reason leading to LAI increase (Zhu et al., 2016), and its contribution (70%) exceeds other factors, such as nitrogen deposition (9%), climate change (8%) and land cover change (4%). The vegetation transpiration feedbacks of elevated CO_2 may act differently among regions due to the characteristics of climate and vegetation cover (Levis et al., 2000). This means that elevated CO_2 can influence runoff signatures through both transpiration (in a direct way) and LAI increasing (in an indirect way). Nevertheless, the separation of the latter from the whole LAI influence is another topic.

The direct impact of elevated CO_2 on transpiration could have noticeable impacts on runoff signatures. One of the advantages of using the improved hydrological models (i.e., PML-V2 model integrated with rainfall-runoff modeling) is that the mechanism of CO_2 impact on stomatal conductance is included (see Gan et al., 2018 and Zhang et al., 2019). Therefore, an additional experiment (Exp.6) is designed using detrended CO_2 as inputs, while other inputs remain dynamic. Then the direct influence of elevated CO_2 on runoff signatures can be isolated by calculating the runoff signature difference between the dynamic experiment and the detrended CO_2 experiment (Exp.1–Exp.6).

To better compare with Section 4.2, Table 5 summarizes the signature changes influenced by LAI increasing and elevated CO_2 using LWE for the last 10 years. The magnitude signatures caused by elevated CO_2 generally increase, and the low flow frequency and duration signatures decrease while high flow ones increase. Same to LAI influences, the dynamic signatures change marginally with almost zero mean values.

In PML-V2, CO_2 is important to constraining canopy conductance and hence it makes CO_2 an important variable for transpiration. Thus, the total runoff would increase with decreased transpiration. This result agrees with the previous argument that elevated CO_2 tends to reduce the stomatal conductance and to weak the transpiration in some regions (Kooperman et al., 2018; Levis et al., 2000; Yang et al., 2019).

Therefore, we can draw a conclusion that the elevated CO_2 acts oppositely comparing to increased LAI, with approaching half of LAI influences on magnitude and dynamic signatures, but less influences on frequency and duration signatures. In other words, the elevated CO_2 may relieve water availability by increasing the runoff amount and decreasing low flow durations. However, the increase in the high flow duration may also lead to extreme floods (Table 5).

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Table 5
A Summary of Runoff Signature Changes Influenced by Increased Leaf Area Index and Elevated CO_2 Using Log-Based Weighted Ensemble for the Last 10 Years

	LAI induced		CO_2 induced	
Runoff signatures	Δsig	Range	Δsig	Range
Qa (mm/yr)	−4.26 ± 1.99	−7.88–−1.62	2.5 ± 1.17	1.21–4.59
Q90 (mm/yr)	−0.02 ± 0.01	−0.04–−0.01	0.02 ± 0.01	0.01–0.03
BFV (mm/yr)	−2.37 ± 0.93	−3.52–−0.9	1.33 ± 0.6	0.58–2.3
Q10 (mm/yr)	−0.002 ± 0.001	−0.005–−0.006	0.001 ± 0.001	0.0004–0.0033
LFD (days/yr)	3.88 ± 3.97	0–11.6	−0.63 ± 0.67	−1.9–0
HFD (days/yr)	−0.42 ± 0.29	−0.9–−0.1	0.17 ± 0.5	0–1.5
MLFD (days/yr)	6.67 ± 14.54	0–45	−6.56 ± 18.57	−56–0
MHFD (days/yr)	−0.56 ± 1.01	−3–0	0.11 ± 0.33	0–1
FDCs (−)	−0.02 ± 0.04	−0.1–0.02	0.01 ± 0.02	−0.02–0.06
CI (−)	0.002 ± 0.003	−0.006–−0.002	0.0008 ± 0.0022	−0.0025–0.0046

Note. Δsig, explained by the mean ± standard deviation, represents runoff signature changes.

5.3. Log-Based Weighted Ensemble Method: A Tradeoff for Proper Runoff Signatures

A wealth of studies claim that multimodel combinations such as the ensemble method are more robust than single modeling (Arsenault et al., 2015; Wan et al., 2021; Zhang, Chen, et al., 2020). The LWE applied in this study is developed based on the works of Dong et al. (2021). This study further improves low flow performance by adding runoff signatures to the calibration scheme and further enlarging the low flow difference with log transformation compared to the original method. Nevertheless, the ensemble size (5 models) was also selected properly to obtain acceptable results, since the increase in ensemble members can obviously enhance the performance of multi-model combinations when the ensemble size is less than six (Wan et al., 2021). Therefore, this method improves the performance of runoff signatures, especially in terms of low flows and overall runoff signatures. However, the pros and cons both exist when applying this method. When reallocating the weighting factors W_i from FDC segments to the runoff series, inconsistency might emerge if the temporal period lags a lot among models. In other words, it is possible that a $W_{i\text{-}low}$ factor of model *A* might encounter $W_{i\text{-}high}$ of model *B* on the same day. The inconsistency unavoidably exists in the reallocated factor matrix, therefore, the performance of the LWE largely relies on temporal predictions among the models. In this context, the improved model should be selected carefully to ensure they provide similar temporal durations. The information in Figure S11 in Supporting Information S1 shows the agreement among LWE and the other seven approaches, indicating the robustness of LWE.

5.4. Potential Hydrological Model Improvements in Terms of Model Structure

In arid and semiarid regions, cease to flow often occurs (Arsenault et al., 2015; Balcombe & Arthington, 2009). Therefore, the subsequent impacts on human livelihood, irrigated agriculture and aquatic organism survival need to be properly predicted. However, the ability to simulate zero flow days is limited for the LWE due to the weighted algorithm and hydrological model structure. Thus, the reduction in water availability is described through annual low flow day (LFD) and annual max low flow duration (MLFD) in this study. Consequently, a method/model for runoff simulation considering runoff signatures, especially for zero flow days and durations, needs to be further developed, which is also a huge challenge for the hydrological community (Bloeschl et al., 2019).

Another model improvement should be vegetation-related conceptual optimization. The advantages of using the PML-V2 model to improve hydrological models in this study are (a) PML-V2 considers interception (strongly related to vegetation canopy) and its evaporation (E_i). (b) PML-V2 incorporates the vapor pressure deficit constraint to constrain canopy conductance and transpiration (E_t). (c) The coupling of the hydrological model and PML-V2 considers soil storage constraining soil evaporation (E_s). (d) the PML-V2 model is feasible in regional estimates (Gan et al., 2021; Zhang et al., 2021), and the PML-V2 product has been widely shown to perform well across the world (McCormick et al., 2021; McNamara et al., 2021; Moletto-Lobos et al., 2020; Zhang

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et al., 2019), indicating a reasonable physics basis. Nevertheless, soil water influences on vegetation dynamics and vegetation change influences on hydrological processes are far more complicated. Vegetation has direct and indirect influences on runoff processes. For direct influences, vegetation changes soil water conservation through root uptake and transpiration. Indirect influences are summarized as follows: (a) the soil structure and attributes (e.g., infiltration rate) can be altered under the influence of roots; (b) the heat flux can also be altered since the canopy contributes greatly to albedo and surface roughness; (c) infiltration might be increased due to leaf litter; and (d) interception strongly relies on the vegetation canopy, thereby reducing throughfall. Moreover, stem flow also alters preferential flow (Crockford & Richardson, 2000). In other words, the vegetation dynamics constrained by soil water and its influences can be better illustrated by improving the model structure considering more detailed physics processes.

This study provides new insights concerning the potential risk to runoff processes due to greening impacts. A more comprehensive development of both modeling and a rich range of study areas should be further conducted in the future. Although it is difficult to obtain relationships between runoff signature change and catchment attributes because of the small size of catchments used, we found that the frequency and duration signatures are more sensitive to LAI increase than other signatures, indicating a higher risk for available water shortages. Based on this work, the future runoff signature change can be estimated using changes in LAI.

6. Conclusions

The impacts of greening on runoff magnitude have been widely investigated, however, few studies have focused on high/low flow frequency, high/low flow duration and dynamic signatures. To this end, this study focuses on the greening impacts on 10 runoff signatures by using five modified hydrological-evapotranspiration models and three ensemble methods. Out of the three ensemble methods, we developed an optimal ensemble method LWE) using a flow-duration-curve-based weighted ensemble strategy, which is better than the other two ensemble methods in simulating runoff time series and runoff signatures. The modeling schemes are tested in the WRB, a major tributary of the Yellow River Basin for the period 2003–2017. We found that for the last 10 years (2008–2017), base flow volume reduction (-2.37 ± 0.93 mm yr^{-1}) makes up approximately 50% of annual runoff change (-4.26 ± 1.99 mm yr^{-1}) with a similar relative change ($\sim -4.5\%$) comparing the two magnitude signatures. Runoff frequency and duration signatures change a few days per year, but with a large range of relative changes (-21.5%–49.5%). Changes in runoff dynamics are small with zero mean values of FDCs and CI. The signature sensitivities to the LAI of the LWE for each basin are presented spatially in absolute and relative forms. When LAI increases by one, the number of annual total low flow days increases by 25.8 ± 25.7 days for each year, the number of annual total high flow days decreases by 2.8 ± 1.4 days for each year, total streamflow reduces by 21.0 ± 4.1 mm/yr and base flow decreases by 12.2 ± 4.7 mm/yr. For each percentage of LAI increase, flow frequency and duration signatures change more rapidly than magnitude signatures ($-0.2 \pm 0.1\%$) in a wider variability ($-0.34 \pm 0.30\%$ for annual high flow days and $0.56 \pm 0.28\%$ for annual low flow days). This study demonstrates that the vegetation greening can result in large impacts on water availability and particularly low flows.

Overall, the WRB, an arid and semiarid sub-basin in the Yellow River Basin with the fastest greening, presents a high risk for available water shortages based on magnitude, frequency and duration signatures. This is especially true for the changes on low flow days. The runoff signature changes quantified here highlight the urgent need for change in water management policy to effectively regulate water resources in the dry basins in China.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The Climate Meteorological Forcing Data set used in this study is publicly available from the National Tibetan Plateau Data Center (https://data.tpdc.ac.cn/en/data/1980e33d-8615-448c-80e3-cfcb635fb110/). The land cover/land use data are provided by the Geographic Data Sharing Infrastructure, Resource and Environment Science and Data Center (https://www.resdc.cn/data.aspx?DATAID=264). The raw LAI, albedo, and emissivity data are obtained from MODIS V6 datasets on the Google Earth Engine (https://developers.google.com/earth-engine/

datasets/catalog/modis, see Table 4 for specific data collections). Global CO_2 data are retrieved from https://gml.noaa.gov/ccgg/trends/gl_data.html.

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